# Worksheet 2.1 — Dividing Polynomials

Grade 12 Advanced Functions (MHF4U) · Unit 2: Polynomial Equations and Inequalities

## The Big Ideas

Long division always works; synthetic division is a shortcut for dividing by $x - a$. Every result is a division statement $P(x) = D(x)Q(x) + R(x)$.

- Long division: divide, multiply, subtract, bring down.
- Synthetic (by $x - a$): bring down, multiply by $a$, add.
- $\deg R < \deg D$; remainder 0 means $D$ is a factor.

## Core Concepts

### The division identity

Polynomial division gives $P(x) = D(x)Q(x) + R$, with the remainder $R$ lower in degree than the divisor $D$.

### Long and synthetic

Long division works for any divisor; **synthetic division** is a fast shortcut when the divisor is $x - a$.

### Fill the gaps

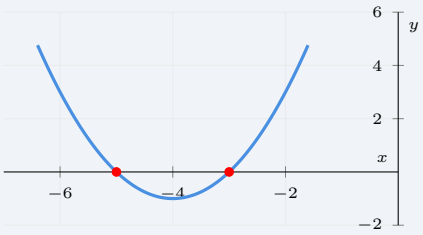
Insert zero coefficients for missing powers so the terms line up correctly through the division.

## Worked Examples

### Example 1 — Exact long division

$(x^2 + 8x + 15) \div (x + 3)$.

***Solution.*** $x^2 \div x = x$; subtract $x^2 + 3x$ $5x + 15$; $5x \div x = 5$; subtract $0$. Quotient $x + 5$, R 0. The graph confirms a zero at $x = -3$:




| | | $x$ | $+5$ |
|---|---|---|---|
| $x + 3$ | $x^2$ | $+8x$ | $+15$ |
| | $x^2$ | $+3x$ | |
| | | $+5x$ | $+15$ |
| | | $+5x$ | $+15$ |
| | | | $0$ |

## Example 2 — Exact long division

$(x^2+7x+10) \div (x+5)$.

***Solution.*** Quotient $x+2$, R 0, so $x^2+7x+10 = (x+5)(x+2)$.

| | | $x$ | $+2$ |
|---|---|---|---|
| $x+5$ | $x^2$ | $+7x$ | $+10$ |
| | $x^2$ | $+5x$ | |
| | | $+2x$ | $+10$ |
| | | $+2x$ | $+10$ |
| | | | $0$ |

## Example 3 — Synthetic division

$(x^3-5x^2+2x+8) \div (x-4)$.

***Solution.*** Coefficients $1, -5, 2, 8$, root 4: bring down 1; $1\cdot 4 = 4$, $-5+4 = -1$; $-1\cdot 4 = -4$, $2-4 = -2$; $-2\cdot 4 = -8$, $8-8 = 0$. Quotient $x^2 - x - 2$, R 0.

| 4 | 1 | $-5$ | 2 | 8 |
|---|---|---|---|---|
| | | 4 | $-4$ | $-8$ |
| | 1 | $-1$ | $-2$ | 0 |

## Example 4 — Division statement

Write the statement for $(x^2+2x-15) \div (x+5)$.

***Solution.*** $x^2+2x-15 = (x+5)(x-3)+0$.

| | | $x$ | $-3$ |
|---|---|---|---|
| $x+5$ | $x^2$ | $+2x$ | $-15$ |
| | $x^2$ | $+5x$ | |
| | | $-3x$ | $-15$ |
| | | $-3x$ | $-15$ |
| | | | $0$ |

## Example 5 — Nonzero remainder

$(x^2+5x+9) \div (x+2)$.

***Solution.*** Quotient $x+3$; $(x+2)(x+3) = x^2+5x+6$; subtract 3. So $x^2+5x+9 = (x+2)(x+3)+3$.

| | | $x$ | $+3$ |
|---|---|---|---|
| $x+2$ | $x^2$ | $+5x$ | $+9$ |
| | $x^2$ | $+2x$ | |
| | | $+3x$ | $+9$ |
| | | $+3x$ | $+6$ |
| | | | $+3$ |

## Example 6 — Synthetic

$(x^3+8)\div(x+2)$.

***Solution.*** Coefficients $1,0,0,8$, root $-2$: $1,-2,4,0$. Quotient $x^2-2x+4$, R 0.

$$\begin{array}{r|cccc} -2 & 1 & 0 & 0 & 8 \\ & & -2 & 4 & -8 \\ \hline & 1 & -2 & 4 & 0 \end{array}$$

## Example 7 — Linear over linear

$(x^2-x-12)\div(x-4)$.

***Solution.*** Quotient $x+3$, R 0, so $(x-4)(x+3)$.

$$\begin{array}{r|cc} & x & +3 \\ \hline x-4 & x^2 & -x & -12 \\ & x^2 & -4x & \\ \hline & & +3x & -12 \\ & & +3x & -12 \\ \hline & & & 0 \end{array}$$

## Example 8 — Remainder

$(x^2+6x+3)\div(x+1)$.

***Solution.*** Quotient $x+5$, remainder $-2$: $x^2+6x+3=(x+1)(x+5)-2$.

$$\begin{array}{r|cc} & x & +5 \\ \hline x+1 & x^2 & +6x & +3 \\ & x^2 & +x & \\ \hline & & +5x & +3 \\ & & +5x & +5 \\ \hline & & & -2 \end{array}$$

## Example 9 — Cubic remainder

$(x^3+3x^2-2)\div(x-1)$.

***Solution.*** Synthetic with root 1 on $1,3,0,-2$: $1,4,4,2$. Quotient $x^2+4x+4$, R 2.

$$\begin{array}{r|cccc} 1 & 1 & 3 & 0 & -2 \\ & & 1 & 4 & 4 \\ \hline & 1 & 4 & 4 & 2 \end{array}$$

## Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

### Question 1

$(x^2 + 10x + 21) \div (x + 3)$?

### Question 2

$(x^2 - 3x - 10) \div (x - 5)$?

### Question 3

Synthetic: $(x^3 - 27) \div (x - 3)$?

### Question 4

$(x^2 + 3x + 7) \div (x + 2)$, with remainder?

### Question 5

Write the division statement for $(x^2 + 8x + 5) \div (x + 3)$.

## Question 6

$(x^2 + 9x + 20) \div (x + 4)$?

## Question 7

$(x^3 - 8) \div (x - 2)$?

## Question 8

$(x^2 - 5x + 6) \div (x - 3)$?

## Question 9

$(x^3 + 4x^2 - 7) \div (x - 2)$, quotient and remainder?

## Question 10

Is $x+5$ a factor of $x^2+3x-10$? (use the remainder)

## Question 11

$(2x^2+7x+3) \div (x+3)$?

## Question 12

$(x^2+1) \div (x+1)$, remainder?

## Question 13 — Challenge

Divide $(x^3-2x^2-5x+6) \div (x-1)$ and state quotient and remainder.

# Question Recap & Answer Key

## Question Recap

1. $(x^2 + 10x + 21) \div (x + 3)$?
2. $(x^2 - 3x - 10) \div (x - 5)$?
3. Synthetic: $(x^3 - 27) \div (x - 3)$?
4. $(x^2 + 3x + 7) \div (x + 2)$, with remainder?
5. Write the division statement for $(x^2+8x+5) \div (x+3)$.
6. $(x^2 + 9x + 20) \div (x + 4)$?
7. $(x^3 - 8) \div (x - 2)$?
8. $(x^2 - 5x + 6) \div (x - 3)$?
9. $(x^3 + 4x^2 - 7) \div (x - 2)$, quotient and remainder?
10. Is $x+5$ a factor of $x^2+3x-10$? (use the remainder)
11. $(2x^2 + 7x + 3) \div (x + 3)$?
12. $(x^2 + 1) \div (x + 1)$, remainder?
13. Divide $(x^3-2x^2-5x+6) \div (x-1)$ and state quotient and remainder.

## ✓ Answer Key

1. $x + 7$, R 0
2. $x + 2$, R 0
3. $x^2 + 3x + 9$, R 0
4. $x + 1$, R 5
5. $(x + 3)(x + 5) - 10$
6. $x + 5$, R 0
7. $x^2 + 2x + 4$, R 0
8. $x - 2$, R 0
9. $x^2 + 6x + 12$, R 17
10. yes (R 0)
11. $2x + 1$, R 0
12. $x - 1$, R 2
13. $x^2 - x - 6$, R 0